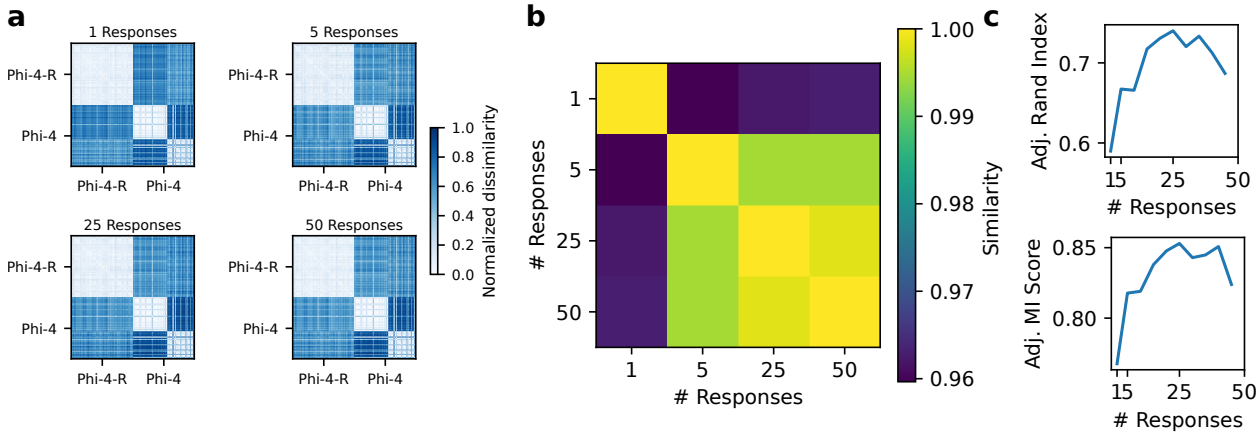

Figure 13: **a** Dissimilarity matrices between the algorithmic fingerprints of different responses, plotted for cluster analyses fitted to different numbers of responses. **b** Similarity between these different dissimilarity matrices. **c** Adjusted Rand Index and Adjusted Mutual Information Score (averaged across 100 responses) for clustering models fit to different numbers of responses.

G.3 Number of samples

To evaluate robustness to sampling choice, we fixed the layer ($L = 17$) and $K = 50$, and evaluated the robustness of clustering to different numbers of responses, between 1 and 50. First, we found very high correlation between assigned clusters across varying number of responses with a minimum Pearson's correlation of 0.96 for using 1 response, and a correlation of 0.9945 between 5 and 25 or 50 responses (Fig. 13a,b). We then measured the consistency of assigned clusters across a varying number of responses with 2 measures: adjusted mutual information (MI) score and adjusted Rand score (Fig. 13c), revealing high consistency for clustering models fit to at least 5 responses. Taken together these findings show robustness of assigned clusters to varying number of responses included in the analysis.